TEAM NUMBER: 65

SwaraAI

Dining out, without needing to see, hear or speak perfectly

Voice-first, adaptive restaurant ordering for Visual, Hearing, Speech and Ageing users in English, Hindi and Telugu, powered by Bhashini.

1 Problem Statement and Our Understanding

A restaurant looks like a simple place, but ordering food there is a **seven-step chain of seeing, hearing and speaking**. It starts with reading a printed menu and ends with splitting a bill. If any one step is out of reach, the whole meal depends on a stranger or a companion. Most AI assistants are designed for a fluent, sighted, hearing, English-speaking user, with accessibility added afterwards. Our brief is to invert that: **start from the person who is excluded at each step, and design every output for them**, in the languages they actually speak (English, Hindi, Telugu).

Where the journey breaks, by condition

Step	Visual	Hearing	Speech	Ageing
1 Read the menu	Printed card, no braille, no audio; phone OCR needs a well-framed photo	Not a barrier	Not a barrier	Small, dense type; multi-column layout
2 Understand dishes	Cannot see veg/non-veg dot or spice icons	Not a barrier	Not a barrier	Culinary jargon (gongura, makhani, tadka)
3 Ask questions	Cannot see the waiter or point at items	Cannot hear the answer	Voice-only assistants fail on atypical speech	Fast speech, long pauses mid-sentence
4 Place the order	Cannot verify what was added	Cannot hear the read-back	Dish names mis-transcribed	Fears a wrong order, long confirmations
5 Call the waiter	Cannot catch attention	Cannot hear "coming"	Cannot call out	Needs repeated, slow requests
6 Understand reply	Fine if spoken	Staff speech is missed entirely	Not a barrier	Noise, accents, speed
7 Pay and split	Cannot read the bill or tax lines	Not a barrier	Cannot explain the split	Mental maths on GST and shares

Why existing tools do not close the gap. Our analysis found seven recurring failures, and each one shapes a design decision:

Failure in typical assistants	What goes wrong at the table	Our design answer
Raw LLM text reaches the user	Markdown, symbols and tables are read aloud as noise, or are invisible to a screen reader	Typed output object plus an **Output Adapter**; no raw text reaches UI or audio
One-size "accessibility mode"	Users are forced to declare a condition, or are locked into a mode that does not fit	Small reversible **feature flags** switched by on-device signals, always with Undo
Literal translation of dish names	"Butter Naan" becomes "butter bread" in Telugu; no waiter recognises it	Protect dish names as tokens, restore them **transliterated** in native script
Numbers and rupees mispronounced	Indic TTS reads "Rs 250" as symbols or skips it	**Number verbalizer** in en, hi, te before TTS
Guesswork on diet and allergies	Jain, vegetarian and nut answers are unsafe if hallucinated	**Rule engine** plus "ask the waiter" escalation; never medical advice
LLM touches money and orders	Wrong totals, silent cart changes, accidental orders	LLM may only call tools; **integer-paise maths** and a spoken confirmation gate
Screen reader talks over audio	TalkBack and app TTS overlap into noise	**Double-speech suppressor** while app audio plays

2 Our Understanding of the Brief and Idea

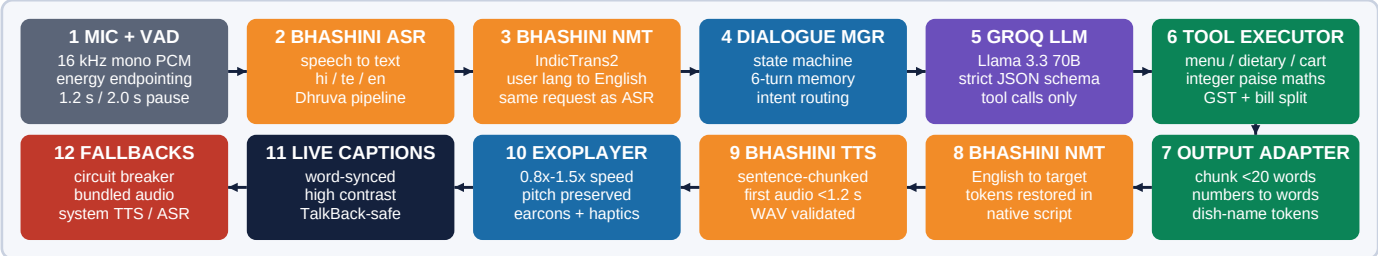
We read the evaluation as three demands: **the output must be accessible** (not only the interface), **the product must adapt to the person** (features, not modes), and **Indian languages must be first-class** (Bhashini as the spine). Dining adds two hard constraints, **safety** (diet, allergens) and **money** (exact totals). **Idea:** SarvaAI takes a diner from photographing a physical menu to paying the bill, and every step can be done eyes-closed. The Visual condition is primary. Hearing, Speech and Ageing are built in depth, not as add-ons.

VISUAL (primary)	HEARING	SPEECH	AGEING
Spoken camera coaching ("too dark", "move phone up"), auto-capture, menu layout narration, voice navigation, TalkBack semantics, earcons and haptics.	Word-highlighted live captions, narrated tables as text too, **Waiter Bridge** that turns staff speech into large-print captions, haptic turn cues.	**Quick Phrase Board** with one-tap spoken requests, longer end-of-speech wait (2.0 s), phonetic matching of mis-heard dish names.	Text to 200%, slower voice (0.8x), three-tier **Explain Simpler**, 64 dp touch targets, repeat on demand, short chunks.

Five design invariants. (1) **Outputs, not just interfaces**: every reply is a typed object adapted per user. (2) **Features, not modes**: a condition inference engine (on at score 70, off at 40, no flapping) toggles single features and announces each change. (3) **Bhashini as spine**: ASR, NMT and TTS for all three languages. (4) **Deterministic actions, probabilistic language**: the LLM interprets, Kotlin executes. (5) **Zero visual dependency**: the full journey works with eyes closed.

3 Bhashini Spine, Models and Pipeline

Every spoken turn is a **speech-to-speech loop** through Bhashini (AnuvaadHub / Dhruva). Orange boxes are Bhashini services. Groq's LLM is used only to understand and explain; all actions run in local code.



Component	Model / service	How we use it, and why
ASR + NMT	Bhashini ASR, then **IndicTrans2** (hi / te to English)	One **chained pipeline request** (ASR then translation) saves a network hop; target under 2.5 s round trip
NMT + TTS	IndicTrans2 (English to hi / te), Bhashini Indic TTS	Reply translated, then synthesised **per sentence** so the first audio starts under 1.2 s while later sentences load
Discovery	getModelsPipeline, then computePipeline	Resolve service IDs and short-lived key; cache 12 h; on 401/403/5xx re-fetch and **replay** the request; probe a **capability matrix** for {en, hi, te} at first launch
Culinary layer	Our wrapper around NMT	Dish names become tokens before translation, then come back **transliterated** (e.g. Butter Naan spoken as batar naan in Telugu script). Numbers are spelled out in the target language first
Playback	Media3 ExoPlayer on device	Speed 0.8x to 1.5x with **pitch preserved** on the client, since server-side speed distorts Indic voices
Vision + dialogue	Groq-hosted Llama (70B dialogue, vision for menu photos)	JSON-schema output at low temperature; menu photo to structured sections, items, prices, veg tags
Waiter Bridge	Bhashini ASR + NMT + TTS across hi / te / en pairs	Diner speaks Telugu, staff hear Hindi; staff reply is captioned in Telugu in large print

4 High-Level Architecture

Layer	Components	Accessibility role
Capture	CameraX, blur / luma / text-clipping analysis, ML Kit	Spoken coaching; auto-shutter after 1.5 s stable; offline OCR fallback
Domain	Room menu store, dietary and allergen rules, cart, GST engine, split-bill, UPI link and QR	Deterministic, unit-testable, conservative on Jain and nut rules
Adaptation	Signal collector (TalkBack, captions, volume, font scale, ASR failures, pauses), hysteresis scorer	Reversible flags with spoken notice and one-tap Undo
Resilience	Network monitor, circuit breaker, bundled en / hi / te audio, system TTS and ASR	Core flows work in airplane mode, with a clear degraded-mode notice

5 Planned Testing Methodology

We test on a **physical Android 16 phone**; emulator-only passes do not count. Fifteen phases are built in order, and a phase must pass its device checks before the next begins.

Layer	What we test	Pass criterion
Unit tests	Number-to-words (en/hi/te), GST rounding, split sums (100 over 3 = 34+33+33), cart maths, dietary rules, fuzzy matcher, hysteresis, malformed-JSON parsing	100% pass; zero rounding drift
Output quality	Raw-vs-accessible comparison view; no markdown or digits reach TTS; sentences under 20 words; native-speaker review of Hindi and Telugu	Reviewer sign-off
Accessibility audit	TalkBack on every screen, Accessibility Scanner, 48 dp+ targets, 7:1 contrast, font scale to 200%, TalkBack/TTS overlap	Zero scanner findings
End-to-end drills	Blindfolded full journey; 70 dB restaurant noise; process kill mid-cart; 30-min memory run; airplane mode; injected 401, 429 and timeouts	No sighted help; state restored; stable memory
Latency and safety	Time to first audio, LLM latency, auto-capture time; fixed set of Jain and allergen questions checked by hand	<3.5 s, <0.6 s, <1.5 s; no unsafe answer

Field validation. Where possible we will run short sessions with blind and elderly volunteers and record task completion, retries and time per step. Failures feed back into phrasing, coaching thresholds and the condition inference weights.